



Technical Service Bulletin: TSB110110

Released Date: 26-Mar-2015

CELECT™ Hard Start

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

The purpose of this topic is to identify some CELECT™ hard starting causes and recommend troubleshooting procedures to identify the problem.

Slow or hard starting (long cranking times) have been investigated on several CELECT™ engines. These engines typically had an 8 to 20 seconds cranking time. All of the hard starting complaints investigated, except for low batteries, have been determined to be caused by no or low cranking fuel pressure, which is created by one of the following conditions:

1. Suction line drain-back.
2. Stuck open pressure regulator

Note : Slow starts is defined as requiring longer than the normal 2 to 5 seconds cranking time.

Hard Start Causes:

- Suction line fuel drain-back to the fuel tank can be caused by:
 - a. Leaking fuel pump front cover seals.
 - b. Rapid restart valve **not** sealing to internal component wear, leaking seals or contamination on the valve seat.
 - c. Any o-ring leaking in the fuel pump, cooling plate or fuel line connections. Reference note below.
 - d. Leaking seat in the gear pump bleed check valve.
 - e. Leaking seat on a Compuchek™ fitting.
 - f. Loose or Leaking Fuel supply hose connections

Note : A small o-ring leak may not have a visible leak but can leak enough to drain the suction line overnight.

- A stuck open pressure regulator will cause fuel pressure to build too slowly. A stuck regulator can be created by any one of the following conditions:
 - a. Contaminants between the plunger and bore.
 - b. Rust on the plunger caused by water ingestion.

- c. Burrs on plunger.
- d. Burrs on the edge of the plunger bore.

Troubleshooting:

- To determine which symptom is causing the hard start, perform the following steps:
 - a. Install a pressure gauge on the Compuchek™ fitting on the side of the shutoff valve. Reference note below.
 - b. Crank the engine for 6 to 10 seconds and observe the pressure gauge while cranking. Reference note below.
 - c. If the pressure starts to move very slowly during cranking, but does **not** reach 25 psi before 6 to 8 seconds, the pressure regulator is sticking open.
 - d. If the pressure gauge does **not** move or there is a pause before moving, during cranking, there is a suction line drain-back.

Note : Do not use any flexible hose on the pressure gauge to prevent slow reaction readings.

- To check for suction line drain-back, perform the following steps:
 - a. Install an 18 inch piece of clear tubing on the suction side of the gear pump or the fuel filter head (Tygon R-3603 is the correct size for use with standard Stratoflex No. 10 hose fittings).
 - b. Start the engine and allow the air to purge from the clear tubing (no bubbles).
 - c. Shut off the engine and observe the clear tubing. Reference note below.
 - d. Observe the column of fuel in the clear tubing, it should stop moving within 1 minute if the system is leak free. If the column continues to move toward the fuel tank, there is a leak in the system. Observe from which direction air bubbles are coming into the clear tubing. If from the fuel tank side, then the leak is in the cooling plate, fittings, fuel line or fuel tank.
 - e. If the fuel column did **not** move immediately, let the truck sit for about 1 hour. If there is no change in the location of the top of the fuel column, the system is sealed and should not cause a hard start due to drain-back of the suction lines.
 - f. Remove the clear tubing, restore the suction line back to its original configuration.

Note : Generally an air space will appear in the clear tubing that is drawn from the top of the fuel filter.

- To check for a sticking pressure regulator perform the following steps:
 - a. Install a pressure gauge on the Compuchek™ fitting in the side of the shutoff valve.
 - b. Start cranking the engine. The pressure gauge should start moving as soon as the engine starts turning over and increase rapidly until the engine fires, at approximately 25 psi, which should take 3 to 5 seconds.
 - c. If the pressure gauge does **not** move or increases very slowly for more than 6 seconds, then the regulator plunger is stuck open and will **not** let pressure build up quickly for a normal start. Reference note below.
 - d. If the plunger is sticking, the gear pump **must** be replaced with a new or ReCon® unit.
 - e. Check the new or ReCon® pressure regulator for sticking before installing the gear pump on the fuel pump body.

- f. Install the fuel pump on the engine, start and purge the air from the fuel lines and shut off the engine.
- g. Restart the engine after letting it sit for at least 30 minutes to check for an improved starting time.

Note : The plunger sticking can be caused by dirt, rust or burrs on the plunger or burrs on the plunger bore.

Document History

Date	Details
2011-4-11	Module Created
2015-3-26	Mispelled word

Last Modified: 26-Mar-2015
